

SangYeon Cho

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Welcome!

"Let's do my best rather than be the best."

Hello, I'm Sangyeon Cho. I am a Ph.D. student at Chung-Ang University, advised by Prof. Junyeong Kim at the IMR Lab, and an AI Researcher at the Korean Surgical Researcher Foundation. My research is on multi-modal learning and medical AI — how to fuse modalities beyond the image-text pair, and how to carry those representations into clinical data, where labels are scarce and the domain gap is wide.

Research Interests

- Multi-modal Learning
- Medical AI

Education

PhD	Chung-Ang Univ , AI Engineering • Advisor: Prof. Junyeong Kim(Labs)	2026.03 – Present
MS	Chung-Ang Univ , AI Engineering • GPA: 4.34/4.5 (Transcripts) • Coursework: Multi-modal Learning, Knowledge Distillation • Advisor: Prof. Junyeong Kim(Labs) • Graduation Thesis	2023.03 – 2025.02
BS	Kookmin Univ , Software Engineering • GPA: 4.09/4.5 (Transcripts) • Coursework: Algorithms, Data Structure, Artificial Intelligence • Graduation Work	2017.03 – 2023.02

Publications

VocMoE: Vocabulary-Guided Mixture-of-Experts for Korean Medical Language Models <i>Sangyeon Cho</i> , Jaeho Han, Mingyu Jeon, Junyeong Kim* Empirical Methods in Natural Language Processing(EMNLP) Findings, 2026	2026.10
VividAC: Visually Informed and Visually Interactive Audio Captioning for Enhancing Audio-Visual Question Answering Mingi Kim, Jaehoon Go, Jinkwon Hwang, <i>Sangyeon Cho</i> , Sunjae Yoon, Junyeong Kim* Interspeech (Oral), 2026	2026.09
Knowledge Beyond Language: Bridging the Gap in Multilingual Machine Unlearning Evaluation Kyomin Hwang, Hyeonjin Kim, <i>Sangyeon Cho</i> , Nojun Kwak* Association for Computational Linguistics(ACL) , 2026	2026.07
Deep Learning-Based Natural Language Processing for Critical Care Identification in Pediatric Emergency Department Jiyoung Agatha Kim, <i>Sangyeon Cho</i> , Minyoung Hwang, Dongjoon Lee, Jangyeong Jang, Rhoan Lee, Changhee Lee, Junyeong Kim, Woori Bae*	2026.03

BMC Emergency Medicine, 2026 [↗](#)

Learning to See through Sound: From VggCaps to Multi2Cap for Richer Automated Audio Captioning 2025.11

Sangyeon Cho, Mingi Kim, Jinkwon Hwang, Jaehoon Go, Minuk Ma, Sunjae Yoon, Junyeong Kim*
[Empirical Methods in Natural Language Processing\(EMNLP\), 2025](#) [↗](#)

Synergy-CLIP: Extending CLIP with Multi-modal Integration for Robust Representation Learning 2025.05

Sangyeon Cho, Jangyeong Jeon, Mingi Kim, Junyeong Kim*
[IEEE ACCESS, Volumn 13, 2025](#) [↗](#)

Improving Multimodal Data Quality with Unified Filtering Score (UF-Score) 2025.02

Sangyeon Cho, Mingi Kim, Hwang JinKwon, Jaehoon Go, Junyeong Kim*
[Association for the Advancement of Artificial Intelligence \(AAAI\) GoodData, 2025](#) [↗](#)

ConCSE: Unified Contrastive Learning and Augmentation for Code-Switched Embeddings 2024.12

Jangyeong Jeon, **Sangyeon Cho**, Minuk Ma, Junyeong Kim*
[IEEE International Conference on Pattern Recognition \(ICPR\)](#) [↗](#)

BioBridge: Unified Bio-Embedding with Bridging Modality in Code-Switched EMR 2024.09

Jangyeong Jeon, **Sangyeon Cho**, Dongjoon Lee, Changhee Lee, Junyeong Kim*
[IEEE ACCESS, Volumn 12, 2024](#) [↗](#)

DSG-KD: Knowledge Distillation from Domain-Specific to General Language Models 2024.08

Sangyeon Cho, Jangyeong Jeon, Dongjoon Lee, Changhee Lee, Junyeong Kim*
[IEEE ACCESS, Volumn 12, 2024](#) [↗](#)

A Pediatric Emergency Prediction Model Using Natural Language Processing in the Pediatric Emergency Department 2024.06

Arum Choi, Chohee Kim, Jisu Ryoo, Jangyeong Jeon, **Sangyeon Cho**, Dongjoon Lee, Junyeong Kim, Changhee Lee*, Woori Bae*
[Nature Scientific Reports, 2024](#) [↗](#)

Experience

Korean Surgical Researcher Foundation, AI Researcher Seoul, Korea
2025.02 – Currently

- K-QIPS Research Analysis
- Korean-style Medical Large Language Model

Lablup Inc., AI Research Assistants Seoul, Korea
2022.06 – 2022.09

- Contribute to the research team
- Developed ground truth data pre-processing module
- Predictive modeling of respiratory infectious disease transmission risk based on ground truth data of droplet spread
- Data augmentation modeling using GAN
- [Seminar LINK](#) [↗](#)

Projects

119 Intelligent report reception voice recognition data construction 2023.09 - 2024.09

- building data pre-processing and analytics pipeline for speech recognition
- modeling dispatch decision-making for decision aids for guard officers
- Tools Used: Pyhon, SQL

Emergency and Critical Patient Classification Using EMR Data

2023.04 - 2024.09

- Designed a model to classify emergency and critical patients using emergency department EMR data.
- Pre-trained the language model using clinical notes collected from real hospitals.
- we additionally utilize clinical note (text) information to utilize the most potential information.
- By combining tabular encoder and text encoder at the embedding level, we achieved better performance than existing models.

Optical Character Recognition (OCR) model improvements

2022.09 - 2023.02

- AI Competition Organized by Software Center University
- Using Pororo as a baseline, we improved it by utilizing data augmentation and adding the Mix-Style module.
- As a result, the team ranked at the top of the leaderboard and won an honorable mention.
- [Presentation LINK](#)

AI speakers for seniors living alone

2022.03 - 2022.06

- Designing an AI speaker that analyzes the emotions of seniors living alone and alerts their caregivers
- Responsible for designing/training deep learning models
- We used two models: Sentiment Classification Model and Sentiment Conversation Model.
- [Github LINK](#)

Research on Meta-Learning Improvement

2022.02 - 2022.12

- Studying/implementing meta-learning in general
- Considered how to improve Model-Agnostic Meta-Learning (MAML) under the guidance of a professor
- Conducted research on applying MAML to natural language processing by combining it with pruning and knowledge distillation
- [Github LINK](#)

Research on Improving Information Security Behavior

2020.10 - 2022.04

- Developing a framework to guide better information security compliance behavior in the enterprise
- Teach principal investigators programming fundamentals
- Teach principal investigators on machine learning and deep learning concepts
- [Github LINK](#)

Awards & Honors

Awards

IEIE Summer Conference, Encouragement Award	2024.06
SW-focused University Collaborative AI Competition, Encouragement Award	2022.09

Honors

Merit Scholarships	2022.09
Merit Scholarships	2022.03
S&T Foundation Scholarships	2022.02

Technologies

Languages: Python, C/C++, C#, Java, SQL, Scala

Frameworks: PyTorch, Hugging Face, TensorFlow

Credentials: ADsP